

Smart Transport Navigation System

1. USER Table

```
CREATE TABLE USER (  
    user_id INT AUTO_INCREMENT PRIMARY KEY,  
    name VARCHAR(100) NOT NULL,  
    email VARCHAR(100) UNIQUE NOT NULL,  
    password VARCHAR(255) NOT NULL,  
    preferred_budget DECIMAL(10,2),  
    preferred_time INT, -- stored in minutes  
    comfort_level INT -- 1 to 5 scale  
);
```

2. LOCATION Table

```
CREATE TABLE LOCATION (  
    location_id INT AUTO_INCREMENT PRIMARY KEY,  
    location_name VARCHAR(255) NOT NULL,  
    latitude DECIMAL(10, 8) NOT NULL,  
    longitude DECIMAL(11, 8) NOT NULL,  
    area_zone VARCHAR(50)  
);
```

3. TRANSPORT_MODE Table

```
CREATE TABLE TRANSPORT_MODE (  
    transport_id INT AUTO_INCREMENT PRIMARY KEY,  
    transport_type VARCHAR(50) NOT NULL,  
    average_speed DECIMAL(5,2), -- km/h  
    base_fare DECIMAL(10,2)  
);
```

4. ROUTE Table

```
CREATE TABLE ROUTE (  
    route_id INT AUTO_INCREMENT PRIMARY KEY,  
    source_location_id INT,  
    destination_location_id INT,  
    total_distance DECIMAL(10,2), -- in km  
    estimated_time INT, -- in minutes  
    estimated_cost DECIMAL(10,2),  
    FOREIGN KEY (source_location_id) REFERENCES LOCATION(location_id),  
    FOREIGN KEY (destination_location_id) REFERENCES LOCATION(location_id)  
);
```

5. ROUTE_SEGMENT Table

```
CREATE TABLE ROUTE_SEGMENT (  
    -- Table definition continues here
```

```

segment_id INT AUTO_INCREMENT PRIMARY KEY,
route_id INT,
transport_id INT,
start_location_id INT,
end_location_id INT,
segment_distance DECIMAL(10,2),
segment_time INT,
segment_cost DECIMAL(10,2),
FOREIGN KEY (route_id) REFERENCES ROUTE(route_id),
FOREIGN KEY (transport_id) REFERENCES TRANSPORT_MODE(transport_id),
FOREIGN KEY (start_location_id) REFERENCES LOCATION(location_id),
FOREIGN KEY (end_location_id) REFERENCES LOCATION(location_id)
);

```

6. TRIP_HISTORY Table

```

CREATE TABLE TRIP_HISTORY (
trip_id INT AUTO_INCREMENT PRIMARY KEY,
user_id INT,
route_id INT,
travel_time INT,
travel_cost DECIMAL(10,2),
trip_date DATETIME,
FOREIGN KEY (user_id) REFERENCES USER(user_id),
FOREIGN KEY (route_id) REFERENCES ROUTE(route_id)
);

```

7. INCIDENT_REPORT Table

```

CREATE TABLE INCIDENT_REPORT (
incident_id INT AUTO_INCREMENT PRIMARY KEY,
user_id INT,
location_id INT,
incident_type VARCHAR(50), -- e.g., 'Flood', 'Accident', 'Protest'
severity VARCHAR(20), -- e.g., 'Low', 'Medium', 'High'
timestamp DATETIME DEFAULT CURRENT_TIMESTAMP,
status VARCHAR(20) DEFAULT 'Active',
FOREIGN KEY (user_id) REFERENCES USER(user_id),
FOREIGN KEY (location_id) REFERENCES LOCATION(location_id)
);

```

8. TRAFFIC_DATA Table

```

CREATE TABLE TRAFFIC_DATA (
traffic_id INT AUTO_INCREMENT PRIMARY KEY,
location_id INT,
congestion_level VARCHAR(20), -- e.g., 'Clear', 'Heavy', 'Gridlock'

```

```
avg_speed DECIMAL(5,2),
date DATE,
time_slot TIME,
FOREIGN KEY (location_id) REFERENCES LOCATION(location_id)
);
```

For: Transport Rating System

```
CREATE TABLE TRANSPORT_REVIEWS (
  review_id INT AUTO_INCREMENT PRIMARY KEY,
  user_id INT,
  transport_id INT,
  rating INT CHECK (rating >= 1 AND rating <= 5),
  comment TEXT,
  timestamp DATETIME DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (user_id) REFERENCES USER(user_id),
  FOREIGN KEY (transport_id) REFERENCES TRANSPORT_MODE(transport_id)
);
```

For: Route Caching Database

```
CREATE TABLE CACHED_ROUTES (
  cache_id INT AUTO_INCREMENT PRIMARY KEY,
  source_location_id INT,
  destination_location_id INT,
  route_data JSON, -- Stores the compiled route steps
  calculated_time DATETIME DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (source_location_id) REFERENCES LOCATION(location_id),
  FOREIGN KEY (destination_location_id) REFERENCES LOCATION(location_id)
);
```

For: Dynamic Fare Management

```
CREATE TABLE TRANSPORT_FARE (
  fare_id INT AUTO_INCREMENT PRIMARY KEY,
  transport_id INT,
  base_fare DECIMAL(10,2),
  cost_per_km DECIMAL(8,4),
  time_multiplier DECIMAL(5,2) DEFAULT 1.0,
  area_zone VARCHAR(50),
  FOREIGN KEY (transport_id) REFERENCES TRANSPORT_MODE(transport_id)
);
```

For: Transport Availability Tracking

```
CREATE TABLE TRANSPORT_AVAILABILITY (
  availability_id INT AUTO_INCREMENT PRIMARY KEY,
```

```

transport_id INT,
location_id INT,
available_count INT DEFAULT 0,
timestamp DATETIME DEFAULT CURRENT_TIMESTAMP,
FOREIGN KEY (transport_id) REFERENCES TRANSPORT_MODE(transport_id),
FOREIGN KEY (location_id) REFERENCES LOCATION(location_id)
);

```

Values of Tables:

1. Insert a User

```

INSERT INTO USER (name, email, password, preferred_budget, preferred_time, comfort_level)
VALUES ('Ahmed', 'ahmed@example.com', 'password123', 200.00, 60, 4);

```

2. Insert Locations (We need at least 5 to match your route query)

```

INSERT INTO LOCATION (location_name, latitude, longitude, area_zone) VALUES
('Mirpur 10', 23.8052, 90.3696, 'North'),
('Banani', 23.7940, 90.4043, 'North'),
('Gulshan 1', 23.7806, 90.4162, 'Central'),
('Dhanmondi 27', 23.7533, 90.3768, 'Central'),
('Motijheel', 23.7289, 90.4189, 'South');

```

3. Insert Transport Modes

```

INSERT INTO TRANSPORT_MODE (transport_type, average_speed, base_fare) VALUES
('Bus', 20.00, 20.00),
('Metro Rail', 45.00, 40.00),
('Rickshaw', 12.00, 30.00);

```

4. Insert Routes (Source 1 to Dest 5, with varying costs and times)

```

INSERT INTO ROUTE (source_location_id, destination_location_id, total_distance,
estimated_time, estimated_cost) VALUES
(1, 5, 12.5, 45, 120.00),
(1, 5, 13.0, 60, 50.00),
(1, 5, 12.0, 30, 200.00);

```

5. Insert Traffic Data (Between 17:00:00 and 18:00:00)

```

INSERT INTO TRAFFIC_DATA (location_id, congestion_level, avg_speed, date, time_slot)
VALUES
(1, 'Heavy', 10.5, '2026-04-19', '17:15:00'),
(1, 'Heavy', 9.0, '2026-04-19', '17:45:00'),
(2, 'Clear', 35.0, '2026-04-19', '17:30:00');

```

6. Insert Incident Report (Active and High Severity)

```
INSERT INTO INCIDENT_REPORT (user_id, location_id, incident_type, severity, timestamp,  
status) VALUES  
(1, 3, 'Waterlogging', 'High', '2026-04-19 16:00:00', 'Active');
```